

tional downloads in the networking subsystem, and creates a structural representation of the document. Modern browsers also contain related subsystems which are used for XHTML, XML and SVG documents.

- ▶ **CSS:** When CSS is encountered, whether that's inside an HTML document or a CSS document, it's passed to a CSS subsystem which parses the style information and creates a structural representation that can be referenced later.
- ▶ **Collections:** HTML documents often contain metadata, for example the information described in the document head or the attributes applied to an element. The collections subsystem is responsible for storing and accessing this metadata.
- ▶ **JavaScript:** When script is encountered, it's passed directly to the JavaScript subsystem which is responsible for executing that script. The JavaScript subsystem is probably the most well-known of the browser subsystems thanks to the visibility it has received over the last few years.
- ▶ **Marshaling:** Because most JavaScript engines are not directly integrated into the browser, there is a communication layer between the browser and the script engine. Passing information through this communication layer is generally referred to as marshaling.
- ▶ **Native OM:** JavaScript interacts with the document through the Document Object Model API's. These API's are generally provided through a subsystem which knows how to access and manipulate the document and is the primary interaction point between the script engine and the browser.
- ▶ **Formatting:** Once the document is constructed, the browser needs to apply the style information before it can be displayed to the user. The formatting subsystem takes the HTML document and applies styles.
- ▶ **Block Building:** CSS is a block based layout system. After the document is styled, the next step is to construct the rectangular blocks that will be displayed to the user. This process determines things like the size of the blocks and is tightly integrated with the next stage - layout.
- ▶ **Layout:** Now that the browser has styled the content and constructed the blocks, it can go through the process of laying out the content. The layout subsystem is responsible for this algorithmically complex process.
- ▶ **Rendering:** The final stage of the process occurs inside the rendering subsystem where the final content is displayed to the user. This process is often referred to as "drawing to the screen" and may occur on the CPU, the GPU, or a combination of both.